



香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

ELEC 2100 Signals and Systems

Wenchao Miao



Outline Chapter 2 Time domain analysis of LTI systems



THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY

1 Overview

2 System Differential Equations (for reference)

3 Continuous-Time System Response

4 Continuous-Time Convolution

5 Difference Equations (for reference)

6 Unit Impulse Response

7 Discrete Convolution

1. Overview of Time Domain Analysis Methods

- **Input/Output Description: Single-variable n-th-order differential (difference) equation**
- **State Variable Description: n-variable first-order differential (difference) equations**

Directly solving system differential/difference equations without transformations offers:

- Intuitive understanding
- Clear physical interpretation
- Foundational basis for learning transform-domain methods

Chapter 2 first introduces the Input/Output description method

1. Overview of Time Domain Analysis Methods

(1) What should we do if the system becomes more complex?

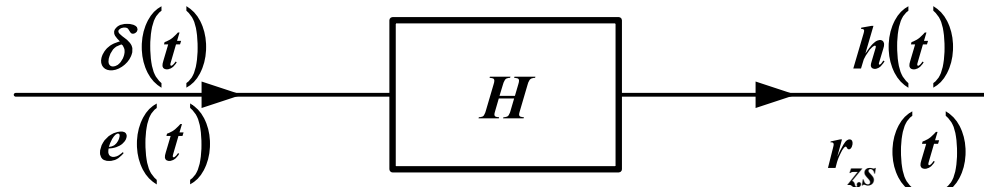
$$\begin{aligned} & a_n \frac{d^n r(t)}{dt^n} + a_{n-1} \frac{d^{n-1} r(t)}{dt^{n-1}} + \dots + a_1 \frac{dr(t)}{dt} + a_0 r(t) \\ &= b_m \frac{d^m e(t)}{dt^m} + b_{m-1} \frac{d^{m-1} e(t)}{dt^{m-1}} + \dots + b_1 \frac{de(t)}{dt} + b_0 e(t) \end{aligned}$$

(2) What should we do if the signal becomes more complex?

- DC, Sine
- Exponential signals, Decaying exponential signals ...
- Random waveforms

1. Overview of Time Domain Analysis Methods

- Classical Method for Solving Continuous-Time Systems
- Impulse response $h(t)$



- **Convolutional** method for **zero-state response** calculation

$$r_{zs}(t) = e(t) * h(t)$$

- Discrete-Time Systems: Analysis methods follow analogous principles

Main Content

1. Description of Linear Time-Invariant Systems
2. Solution Methods for System Differential Equations
3. Classical Method

Complete Solution = **Homogeneous** Solution + **Particular** Solution

2. System Differential Equations (for reference)

1. Description of Linear Time-Invariant Systems

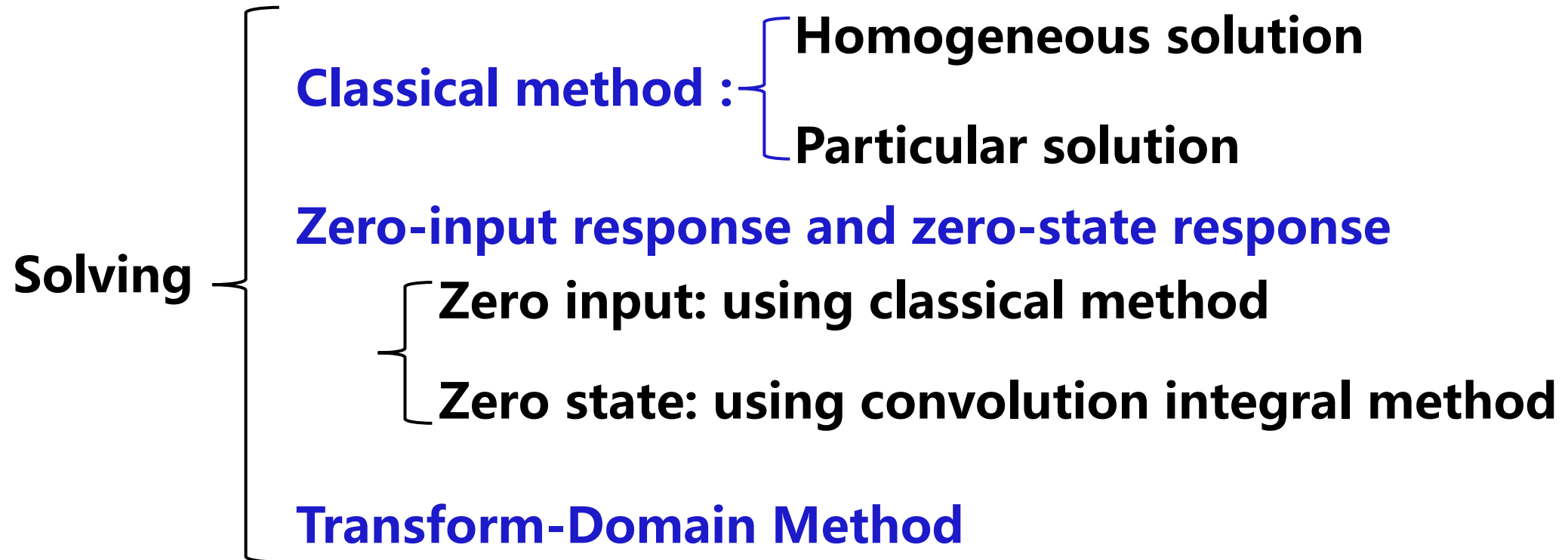
For a **LTI system**, the relationship between the excitation signal $e(t)$ and the response signal $r(t)$ can be described by the following **differential equation**

$$\begin{aligned} & a_n \frac{d^n r(t)}{dt^n} + a_{n-1} \frac{d^{n-1} r(t)}{dt^{n-1}} + \cdots + a_1 \frac{dr(t)}{dt} + a_0 r(t) \\ &= b_m \frac{d^m e(t)}{dt^m} + b_{m-1} \frac{d^{m-1} e(t)}{dt^{m-1}} + \cdots + b_1 \frac{de(t)}{dt} + b_0 e(t) \end{aligned}$$

n, m is an integer, and the coefficients are all constants

System Order: Determined by **the highest differential term**

2. Solution Methods for System Differential Equations



Summary

Classical method for solving differential equations:

- Complete solution = Homogeneous solution + Particular solution
- **Homogeneous solution method** :
Characteristic Equation $\rightarrow$ Characteristic Root $\rightarrow$ Homogeneous Solution
- **Particular solution method**
- The particular solution is in the same form as the input
- Undetermined coefficient: substituting it into the original equation
- **Determination of the coefficient of homogeneous solution**: using the Initial condition of the system

Main Content

1. Classification Methods for System Responses
2. Impulse Response and Step Response of Continuous-Time Systems

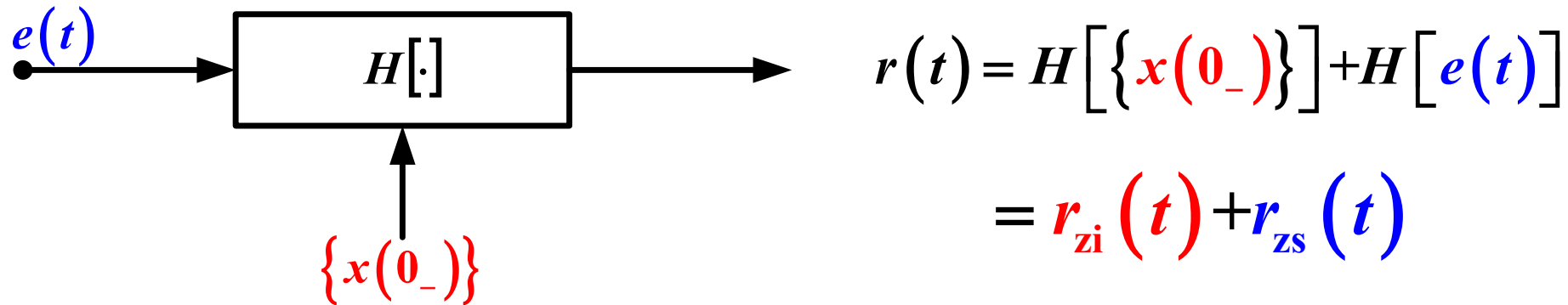
3. Continuous-Time System Response

1. Classification Methods for System Responses

- **Zero-input + Zero-state**
- **Natural + Forced**
- **Transient + Steady-state**

3. Continuous-Time System Response

1.1 Zero-input + Zero-state



(1) Zero-Input response (set input to zero)

Definition: Response produced exclusively by the system's initial energy storage (initial state)

Mathematics: Solution to the homogeneous differential equation with initial conditions

(2) Zero-State response (set the initial state to zero, no initial energy storage)

Definition: Response produced exclusively by the external excitation signal

Mathematics: Convolution of input with system's impulse response

(1) Solution of Zero-Input response

$$r_{zi}(t)$$

Input : 0

Equation : Homogeneous equation

Solution Form: Particular solution is zero, only homogeneous solution

Boundary conditions (without jumping) :

$$r_{zi}(0_-) = r(0_-), r_{zi}'(0_-) = r'(0_-), \dots, r_{zi}^{(n-1)}(0_-) = r^{(n-1)}(0_-)$$

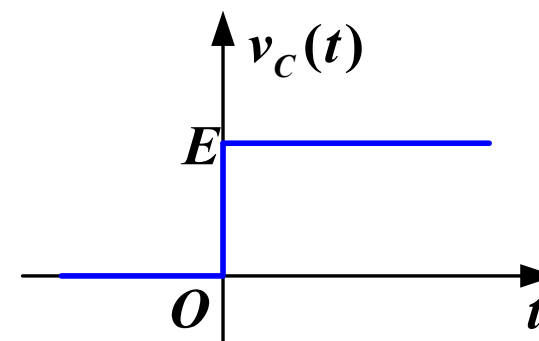
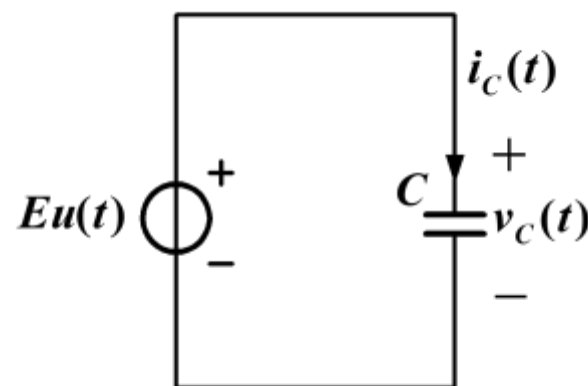
3. Continuous-Time System Response

(2) Jump Discontinuity at the Initial Point

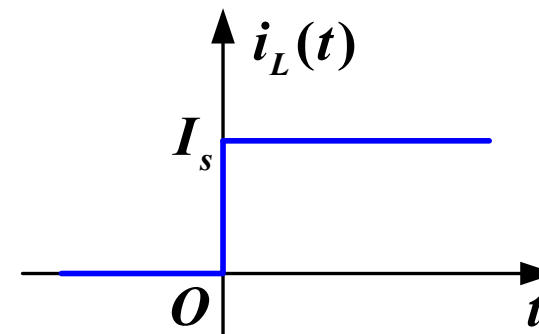
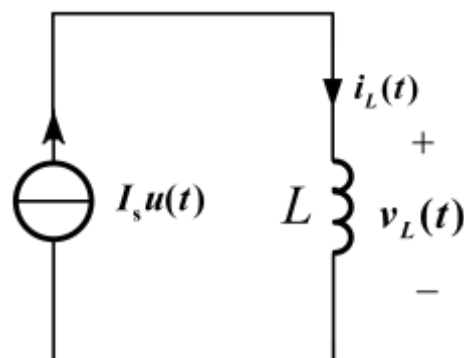
$$v_C(0_-) = 0$$

$$v_C(0_+) = E$$

$$i_C(t) = C \frac{dv_C(t)}{dt}$$
$$= CE\delta(t)$$



$$v_L(t) = L \frac{di_L(t)}{dt}$$
$$= LI_s\delta(t)$$



A sudden change in the system's state variables (e.g., capacitor voltage, inductor current) at the moment of excitation ($t = 0_+$), violating continuity conditions

3. Continuous-Time System Response

(3) Solution of Zero-state response

$$r_{zs}(t)$$

Zero State: $r_{zs}(0_-) = r'_{zs}(0_-) = \dots = r_{zs}^{(n-1)}(0_-) = 0$

Input : Given

Equation : Original

Solution Form: Particular solution + Homogeneous solution

Jumping: $r_{zs}(0_+), r'_{zs}(0_+), \dots, r_{zs}^{(n-1)}(0_+)$

Solution is complex, hence the convolution integral method is introduced

(4) Significance of Zero-State Response Analysis

- For most practical communication and electronic systems, only the **zero-state response** needs to be studied
- The zero-state response can be solved using the **convolution method**
- System "**identification**" and analysis of its inherent characteristics can be achieved through input-output relationships

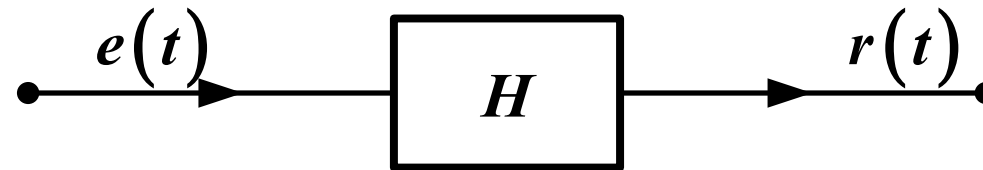
3. Continuous-Time System Response

2. Impulse Response and Step Response

- **Definition of Impulse response and Step response**
- **The impulse response of the typical systems**
- **Using Impulse response to judge the causality and stability of a system**

2.1 Definition of Impulse response and Step response

(1) Unit Impulse response



The zero-state response of a system excited by a unit impulse signal $\delta(t)$ is called the **unit impulse response** (or simply impulse response), denoted as $h(t)$

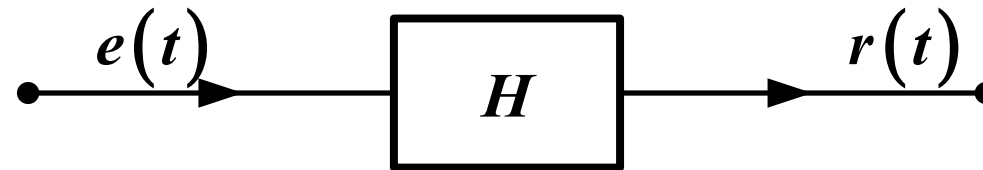
$$h^{(k)}(0_-) = 0 \quad (k = 0, 1, \dots, n-1)$$

$$\delta(t) \xrightarrow{\text{zs}} h(t)$$

3. Continuous-Time System Response

2.1 Definition of Impulse response and Step response

(2) Unit step response



The zero-state response of a system excited by a unit step signal $u(t)$ is called the unit step response (or simply step response)

$$g^{(k)}(0_-) = 0 \quad (k = 0, 1, \dots, n-1)$$

$$u(t) \xrightarrow{\text{zs}} g(t)$$

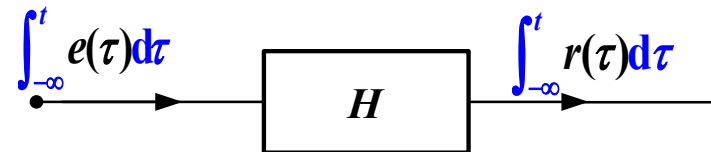
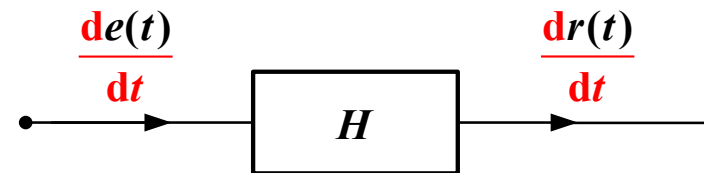
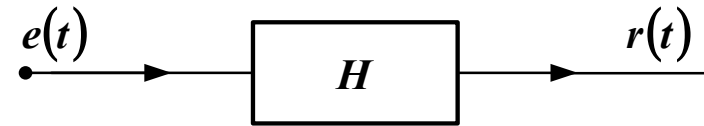
3. Continuous-Time System Response

2.1 Definition of Impulse response and Step response

(3) Impulse response vs Step response

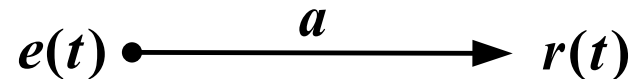
$$u(t) = \int_{-\infty}^t \delta(\tau) d\tau$$

$$g(t) = \int_{-\infty}^t h(\tau) d\tau$$



2.2 The impulse response of the typical systems

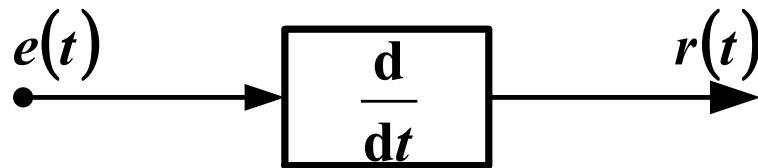
(1) Scalar multiplier (Digital multiplier, scaler)



$$r(t) = ae(t)$$

$$h(t) = a\delta(t)$$

(2) Differentiator

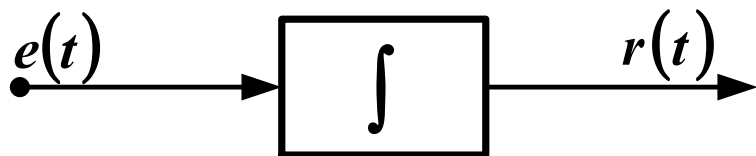


$$r(t) = \frac{de(t)}{dt}$$

$$h(t) = \frac{d\delta(t)}{dt} = \delta'(t)$$

2.2 The impulse response of the typical systems

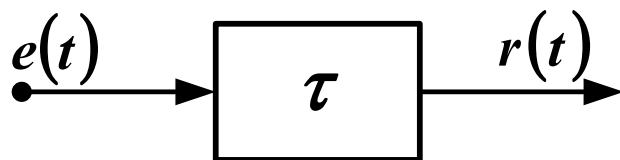
(3) Integrator



$$r(t) = \int_{-\infty}^t e(\tau) d\tau$$

$$h(t) = \int_{-\infty}^t \delta(\tau) d\tau$$
$$= u(t)$$

(4) Delayer (Unit Delay Element)



$$r(t) = e(t - \tau)$$

$$h(t) = \delta(t - \tau)$$

2.3 To judge the causality and stability of a system

The unit Impulse response of the causal system satisfies the following properties

$$h(t)=0 \quad (t < 0)$$

Applying the causality concept to signals, a signal is **causal**:

$$x(t) = 0 \quad (t < 0)$$

The unit impulse response of a **stable system** satisfies the following properties

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty$$

Summary

1. Classification Methods for System Responses

2. Impulse Response and Step Response of Continuous-Time Systems

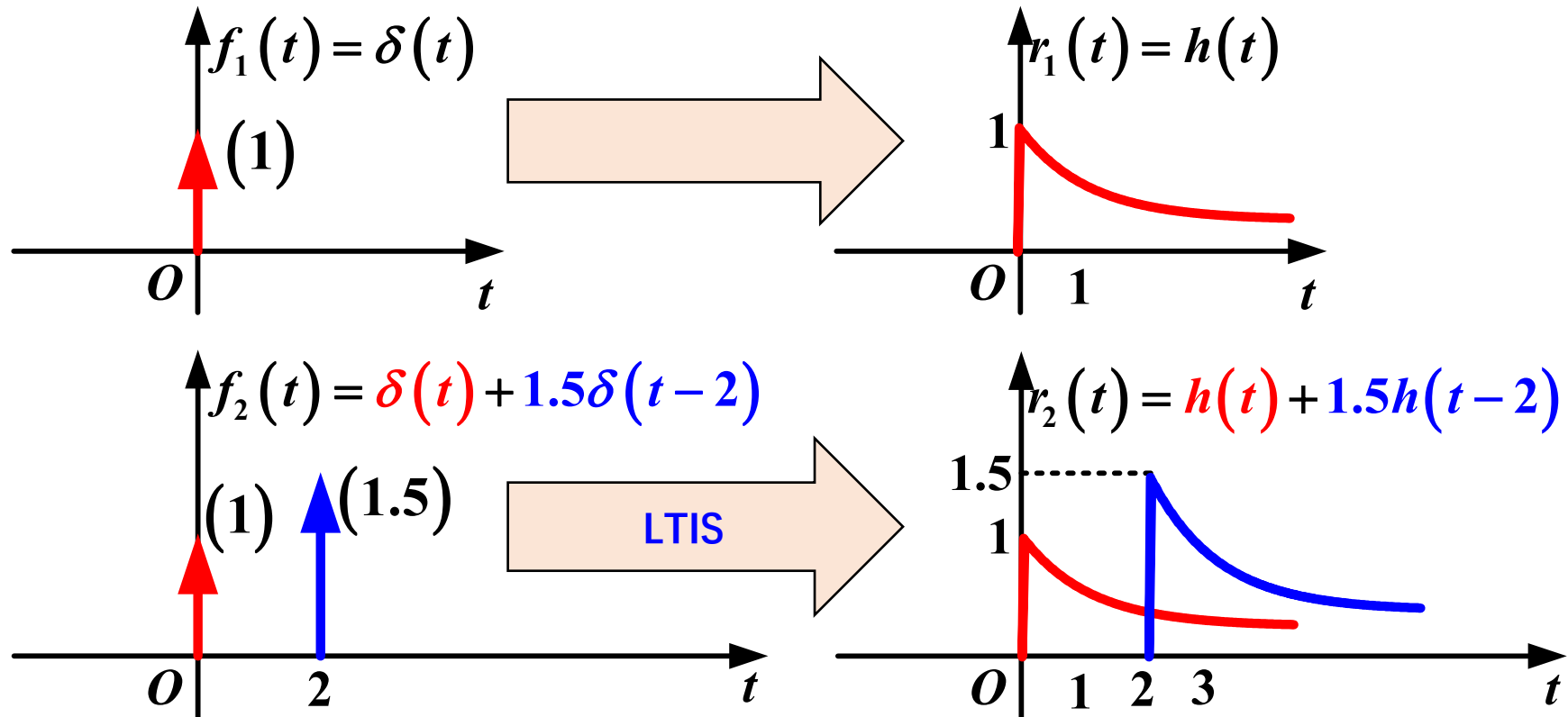
- Definition of Impulse response: **The zero-state response of a system under the action of a unit impulse signal $\delta(t)$ is called the impulse response**
- The impulse response can fully characterize system behavior
- Three time-domain solution methods for determining impulse response: (1) Singularity function balance method, (2) Time-Domain Direct Solution, (3) Convolution Integral
- The time-domain method is physically intuitive, but the calculation is cumbersome
- Calculation of impulse response and step response is simpler and more convenient using transform-domain methods (e.g., **Laplace transform**)

Main Content

1. Introduction to Convolution
2. Zero-State Response via Convolution
3. Physical Interpretation of Convolution
4. Properties of Convolution
5. Convolution Calculation

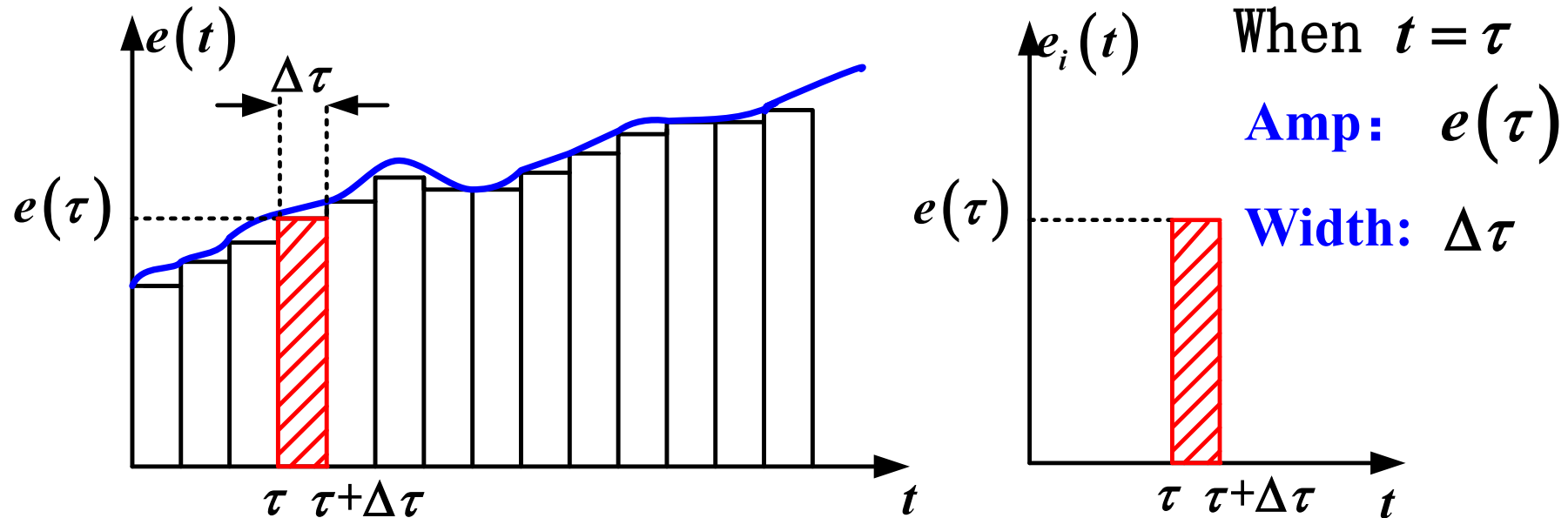
4. Continuous-Time Convolution

1. Introduction to Convolution



How to determine the **zero-state response** of a random signal $f(t)$ passing through the system?

1. Introduction to Convolution



Narrow pulse signals can be represented as

$$e_i(t) = e(\tau) [u(t - \tau) - u(t - \tau - \Delta\tau)]$$

From $\tau = -\infty$ to ∞ , $e(t)$ can be represented as the superposition of many narrow pulses

$$e(t) \approx \sum_{\tau=-\infty}^{\infty} e(\tau) [u(t - \tau) - u(t - \tau - \Delta\tau)]$$

1. Introduction to Convolution

$$e(t) \approx \sum_{\tau=-\infty}^{\infty} e(\tau) \frac{[u(t-\tau) - u(t-\tau-\Delta\tau)]}{\Delta\tau} \cdot \Delta\tau$$

Let $\Delta\tau \rightarrow 0$

$$\Delta\tau \rightarrow d\tau$$

$$\sum_{\tau=-\infty}^{\infty} \rightarrow \int_{\tau=-\infty}^{\infty}$$

$$\lim_{\Delta\tau \rightarrow 0} \frac{[u(t-\tau) - u(t-\tau-\Delta\tau)]}{\Delta\tau} = \frac{du(t-\tau)}{dt} = \delta(t-\tau)$$

difference

differential

Represent **any** signal as a **weighted superposition** of **time-shifted unit impulse signals**

$$e(t) = \int_{-\infty}^{\infty} e(\tau) \delta(t-\tau) d\tau$$

2. Zero-State Response via Convolution

A mathematical operation that combines two functions (typically a signal and a system's impulse response) to produce a third function representing how one modifies the other

A signal $e(t)$ can be represented as the sum of impulse sequences

$$e(t) = \int_{-\infty}^{\infty} e(\tau) \delta(t - \tau) d\tau$$

If it is applied to LTIS with an impulse response of $h(t)$, the response is

$$\delta(t) \rightarrow h(t)$$

Time-Invariance

Linearity

$$\delta(t - \tau) \rightarrow h(t - \tau)$$

$$e(\tau) \delta(t - \tau) \rightarrow e(\tau) h(t - \tau)$$

Zero-state
response

$$e(t) = \int_{-\infty}^{\infty} e(\tau) \delta(t - \tau) d\tau \rightarrow \int_{-\infty}^{\infty} e(\tau) h(t - \tau) d\tau = r(t)$$

2. Zero-State Response via Convolution

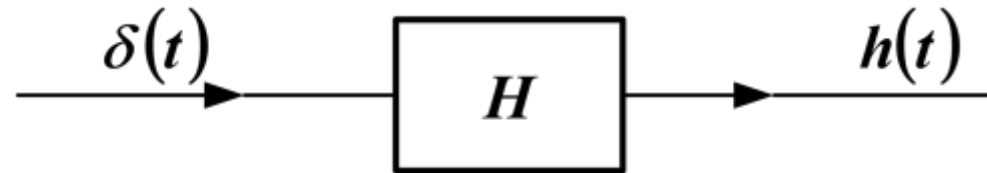
With two functions $f_1(t)$ and $f_2(t)$, integral

$$f(t) = \int_{-\infty}^{\infty} f_1(\tau) f_2(t - \tau) d\tau$$

Convolution integral of $f_1(t)$ and $f_2(t)$, abbreviated as **convolution**, denoted as

$$f(t) = f_1(t) \otimes f_2(t) \quad \text{or} \quad f(t) = f_1(t) * f_2(t)$$

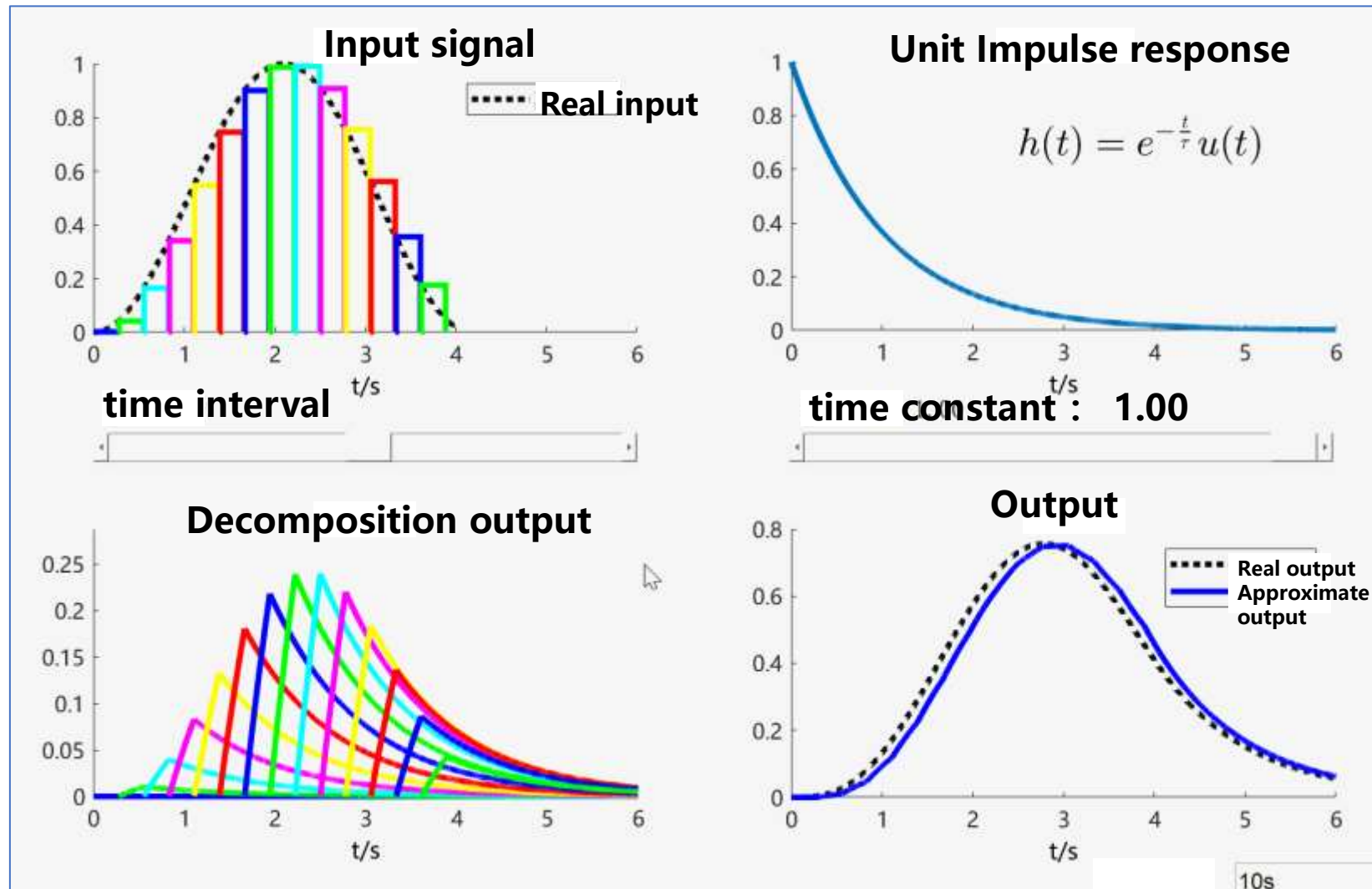
The zero-state response of the system can be solved using convolution:



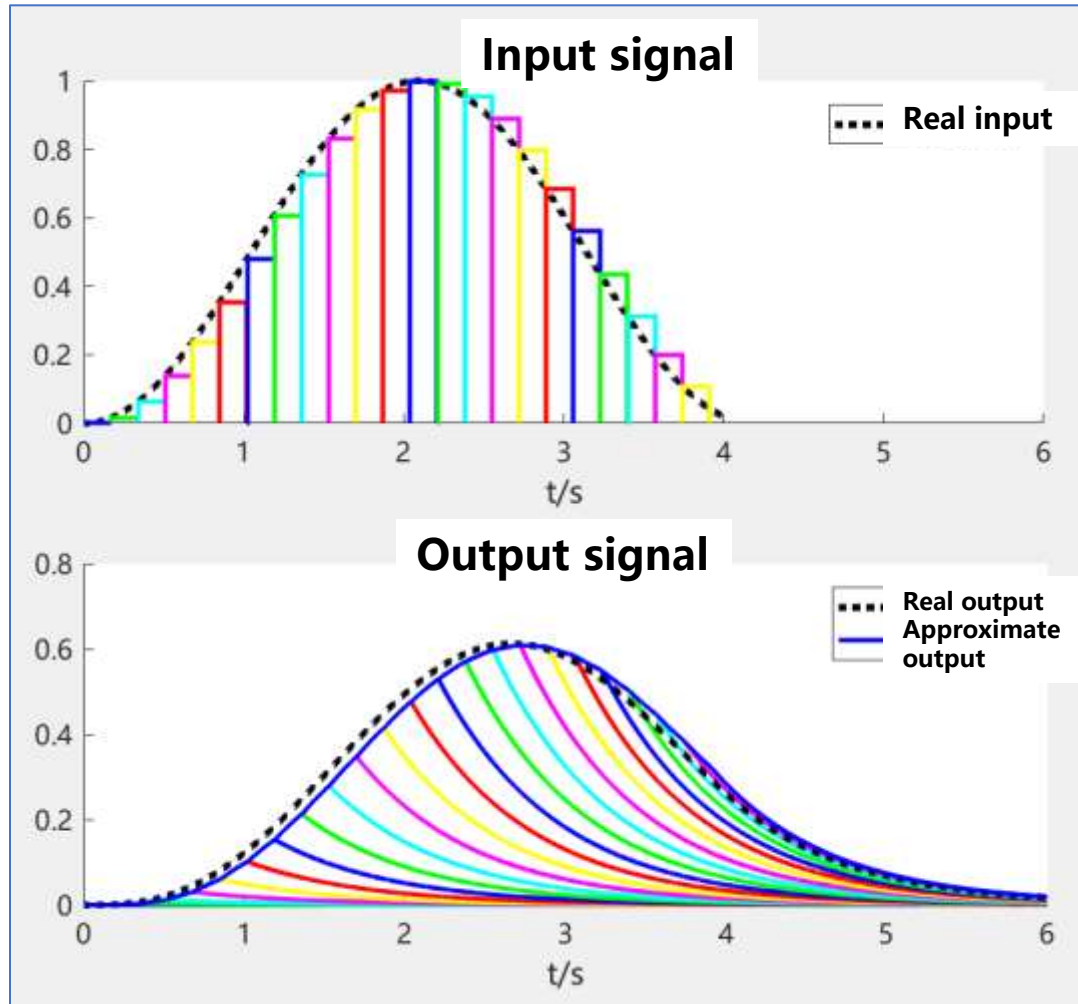
$$r_{zs}(t) = \int_{-\infty}^{\infty} e(\tau) h(t - \tau) d\tau$$

4. Continuous-Time Convolution

3. Physical Interpretation of Convolution



3. Physical Interpretation of Convolution



$$r_{zs}(t) = \int_{-\infty}^{\infty} e(\tau) h(t - \tau) d\tau$$

τ : The moment when the signal occurs

$e(\tau)$: excitation at time t

t : Time to observe the response

$h(t - \tau)$: The system's impulse response at time t due to a unit impulse applied at τ